A Perspective into the Interactions with Musical Artificial Intelligence as a Partner, Product, and Tool

First Author[1] , **Second Author**[2] , **Third Author**[2,3] and **Fourth Author**[4]
[1]First Affiliation
[2]Second Affiliation
[3]Third Affiliation
[4]Fourth Affiliation
{first, second}@example.com, third@other.example.com, fourth@example.com

Abstract

Independent from its technical details, a musical Artificial Intelligence system with unsupervised learning can have various roles in interactive artworks, musical performances, and generative music systems. Although there is a body of exciting recent advancements in AI systems for musical applications, the role of these systems in musical practices are still to be manifested. In this paper, we present ArchXXX which is a Machine Learning pipeline that combines a self-organizing map with three statistical sequence modelling algorithms. The architecture utilizes a dataset of audio recordings, which is analyzed through a machine listening pipeline. The machine listening segments and thumbnails the audio data to automatically create a dataset of audio segments. Ideally, these segments are snippets of sonic gestures. The audio segments are clustered within a latent space of discrete clusters using a Self-Organizing Map. The ML pipeline in ArchXXX utilizes the latent audio space to create a custom symbolic representation of audio recordings and musical compositions. These sequences are modelled through statistical sequence modelling algorithms to capture the musical form. This combination of latent sound space with the temporal sequence modelling creates affordances to incorporate ArchXXX into a variety of musical applications. We explore three cases where ArchXXX is incorporated into three artworks, as a partner, a product, and a tool.

1 Introduction

The advancements in Machine Learning and Artificial Intelligence initiated many breakthroughs in a variety of disciplines, and many breakthroughs also have appeared in Arts and Music. Machine Learning and Artificial Intelligence architectures have been applied to musical practices, applications, and scenarios. The technology contributions of AI have been noticeable in music technology research, which evolved in to the emergence of Musical Artificial Intelligence.

The applications of musical AI spans some variety of musical practices. There has been a several of literature reviews on Musical AI systems from a variety of perspectives. Carnovalini and Rodà (2020) presents a survey of literature reviews of Musical AI systems which includes seven distinct papers found by searching terms "computational creativity music," "musical metacreation," "algorithmic composition," and "music generation" followed by "review" or "survey". This meta-review covers reviews in English published between 2010–2019, and we use the same approach in our coverage of musical AI surveys below. Nierhaus's book titled *Algorithmic Composition* (2009) is a broad coverage of all algorithmic approaches in musical practices. Building on the foundation of Algorithmic Music, the survey by Fernández Rodríguez and Vico Vela (2013) specifically concentrates on Artificial Intelligence in Music while incorporating the research in Computational Creativity into their analysis. This review covers the AI systems for generating symbolic music in composition practices. The Musical AI systems review by Williams *et al.* (2015) introduces a novel perspective by looking into the Affective Computing in Musical AI systems. Affective Computing is a discipline where computational approaches are developed to estimate emotional aspects of a content such as image, video, sound, or music. Similar to Williams *et al.*, the taxonomy proposed by Herremans *et al.* (2017) mentions a specific affect, perceived emotion as a property connecting an item called *narrative* to another item called *composition* in their concept map of musical AI systems. The following two surveys start from a specific AI algorithm type and then expands into a variety of dimensions including musical scenarios in their taxonomy. Starting from the literature of multi-agent systems in AI, Tatar and Pasquier (2019) delves into seventy-eight musical AI systems in their literature review. Their taxonomy emerges from those systems, and mainly grounded in the multi-agent system architectures and how different ML and AI algorithms fit within an umbrella of multi-agent systems. While the survey by Tatar and Pasquier has foundations in multi-agent systems, [Briot *et al.*, 2017] surveys musical AI systems that utilizes Deep Learning architectures in a variety of musical applications.

In this work, we built on this literature of musical AI and present a Machine Learning (ML) pipeline that brings in intriguing interaction varieties. The technical discussions in this paper covers the details of how the pipeline brings together multiple ML algorithms together to create a musical AI system with capability of aesthetic diversity. The technical explanations in this paper serves the investigations of how this musical AI architecture can be manifested in differ-

ent ways as a partner, product, and a tool. To discover these three emerging roles in human-machine interactions within musical AI in artistic practices, we bring in the methodologies that are known to the third wave HCI studies[Bødker, 2006; Bødker, 2015; Filimowicz and Tzankova, 2018]. These methodologies emphasize first-person experiences in human-machine interaction through post-phenomenology lenses, and how such investigations combined with thematic analysis can help the understanding, conceptualization, categorization, and communication of new and upcoming technologies, specifically musical AI systems.

2 Methodology

In this section we outline methodological and analytical approaches that helped us to contextualize the recurring paradigms in the artistic manifestation and utilization of the musical AI system.

2.1 First person experience and post-phenomenology as Design Method

The increasingly complex navigation of human-agent interaction within musical and artistic contexts has seen the deployment of methodologies and analytical approaches which addresses the subjective nature of artistic research. One such methodological approach is post-phenomenology, utilised as a reflective lens in the analysis of a human and technology interaction. Post-phenomenology is a philosophical approach in the study of technology, specifically examining the interaction between a human actor and technology (or technological system) to discern how each shape each other [Ihde, 1990; Verbeek, 2008; Rosenberger and Verbeek, 2015]. A central theoretical concept within a post-phenomenological enquiry is Ihde's analytical framework- constituting examination of interaction through four human-technology relations: hermeneutic, embodiment, alterity and background.

The reflection of how these four relations are enacted by both the human actor and agent system within the context of a musical interaction may offer deeper insights into how these these relations are formed throughout a design process, and further- the nature of the musical 'role' that the technology is performing with the human actor within the interaction.

Essential to this understanding of human-technology relations is the recognition of first-person experiences as a valuable material in a design process. The generative nature of first-person experience and perspectives as a resource or tool for design has seen an increasing abundance of research work spanning the Arts and HCI, examining the [McCarthy and Wright, 2007; Lucero *et al.*, 2021; Svanæs and Barkhuus, 2020; Varela and Shear, 1999; Cross, 2006; Höök *et al.*, 2021; Desjardins *et al.*, 2021]. A first-person research methodology takes the researcher's firsthand and subjective experience as the point-of-departure for generating knowledge and as a resource for design [Ellis and Bochner, 2000; Neustaedter and Sengers, 2012]. The utilisation of first-person experience as design material become especially pertinent in the later explanation (see Section 4 of the significance of the first author's utilisation–and familiarity with–of self-produced sonic material in informing subsequent design iterations pertaining to audio feature selection.

2.2 Research-through-Design

Offering a process-oriented approach to knowledge development, Research-through-Design (RtD) is a methodological approach which foregrounds the design process itself as a mechanism for the formation of knowledge. Coined as a term by [Frayling, 1994], RtD is an approach that considers the practitioner's accumulative insights during the course of their engagement in a design practice, and frames their iterative experimentation with new processes and materials as a mechanism for generating knowledge [Zimmerman *et al.*, 2007]. As a method of inquiry that centres exploration and experimentation, RtD has been highly recognised and utilised approach within fields such as Interaction Design and HCI [Redström, 2017; Krogh and Koskinen, 2015]. Accordingly, the examination of the first author's design iterations and experimentation throughout their system development and utilisation of ArchXXX across varying musical settings reveals procedural thinking corresponding with RtD process outcomes.

2.3 Thematic Analysis

An additional approach relevant to this paper's inquiry is the use of thematic analysis to examine and reflect upon emergent patterns resulting from the production of a design artefact. Discussion on the use of thematic analysis as a vehicle for identifying and interpreting patterns in data, as illustrated by [Clarke and Braun, 2017], offer a qualitative-centric approach to reflection. The flexibility of this analytical approach has seen its implementation in research work that affords space for the researcher's own subjectivity, interpretation and experiences in understanding the XYZ. [Kiger and Varpio, 2020] This is especially applicable to the presented case study in Section 4, wherein the analysis of how the ArchXXX architectures are utilised in varying musical contexts reveal framings of the system as constituting a musical *partner*, *product*, or *tool* usage.

2.4 Musical AI as a Tool, a Product, or a Partner

Many taxonomies, categorizations, and classifications of Musical Machine Learning and Artificial Intelligence situates their foundation of the specifics of technology Carnovalini and Rodà. Recent developments in ML and AI showed that technology advancements are more rapid then ever. As the technology transforms and shifts, many of the taxonomies fall short in addressing novel and emerging possibilities in technology advancements. In this work, we propose three main categories of Musical AI interactions where we chose a technology agnostic direction.These categories emerged from the interaction situations between the human and the machine in musical scenarios, in close connections with the utilization and the manifestation of the technology by the artist.

3 Case Study: ArchXXX

ArchXXX[1] is a family of agent architectures for interactive music system applications such as autonomous composing, machine improvisation, and audio-based interactive multimedia systems. ArchXXX architectures apply unsupervised learning on audio recordings; hence, the agent architecture is

[1]The open-source code is available at the following link: https://anonymous.4open.science/r/mlaudiosalad-2644

flexible to be trained on various musical styles. The shared property of ArchXXX architectures is the combination of the latent sonic space generation algorithm with statistical sequence models. The sonic latent space generation uses an unsupervised, fully-connected neural network algorithm called Self-Organizing Maps (SOMs). SOMs have been applied to various Machine Learning (ML) tasks such as dimensionality reduction, clustering, topology modelling, and latent space generation. While the SOM in ArchXXX organizes the sound memory in the latent sonic space, the statistical sequence models organize the sound object selections in time. The sequence modelling algorithms are trained on the sound clusters that are generated by the SOM. The sequence modelling algorithms handle the musical temporality and model the musical structure.

3.1 System Architecture

The architecture of ArchXXX creates a sound memory through automatic audio segmentation and thumbnailing, using audio features of timbre, loudness, fundamental frequency, duration, and sound affect features of eventfulness and pleasantness. The architecture applies Self-Organizing Maps [Kohonen, 1982; Kohonen, 1998], a neural network machine learning algorithm. Using the SOM, the agent organizes sounds on a two-dimensional map so that similar sound clusters locate closer to each other. ArchXXX learns the temporality of musical form by applying pattern recognition on the organized sound memory. The agent assumes the musical form as temporal shifts on sound clusters that are organized in the feature space.

Machine Listening

The machine listening module consists of two steps: segmentation and thumbnailing. The *segmentation* module in ArchXXX applies a novelty based onset detection using FFTs. We use MAX MuBu[2] external library [Schnell *et al.*, 2009] for the segmentation and the audio feature extraction. The MuBu library includes the IRCAM descriptors audio feature extraction library and Plug-In-Plug-Out (PIPO) [3] signal processing tools. The hybrid corpus of the agent consists of both audio segments and their thumbnails represented as a 35-dimensional vector. This vector includes statistics (mean and standard deviation) of audio features and two additional sound affect features. The sound affect estimation in ArchXXX is a multivariate regression model, and the details of this ML model can be found in (CITEY).

In the following, we delve into clustering the audio segments using the selected audio feature labels.

Organizing Sounds with Self-Organizing Maps

Self-Organizing Map (SOM) is an artificial neural network (ANN) for dimensionality reduction, clustering, and visualization applications. SOMs apply unsupervised learning to automatically generate a 2-dimensional symbolic representation of data. Common topologies of SOMs are square, rectangular, and toroid shapes. The map consists of a finite number of nodes. A node is a vector with the same number of dimension as the input data. In SOM, similar nodes locate closer to each other. Therefore, SOMs take the shape of input topology and can show the similarities in the input data.

The training procedure of a SOM passes the training dataset more than once. Each pass is called an *epoch*. It is common to decrease the learning rate and the neighbouring radius as the epoch number increases. Because of this decrease in learning rate and the neighbouring radius, SOMs can capture the generalised topology as well as the particularities of the training data. After the training, we can use an SOM to cluster the input data. Each node in a SOM represents one cluster. The clustering procedure applies a distance measure to find the closest SOM node of an input vector (n_c). Using a trained SOM, we can cluster both the training data or new data.

In ArchXXX's learning, the topology is a square of $a * a$. We aim for 6 samples per node in average. In our trials, aiming 6 samples per cluster gave a low number of empty clusters. ArchXXX uses the following formula to calculate the parameter a:

$$a = int(\sqrt{\text{the number of audio samples}/6}) \quad (1)$$

The total number of epochs is 1000. The learning rate is linearly decreasing from 0.25 to 0.01 with each epoch. The neighbourhood radius is also linearly decreasing from $a/4$ to 0. For a given audio corpus, we calculate the mean and the standard deviation of each audio feature, and use these statistics to standardize the SOM input vectors:

$$I_{norm}[i] = \frac{I[i]}{M[i]} * STD[i] \quad (2)$$

and $i \in [1 : N]$, where the N is the total number of dimensions of the input vector, I is the calculated audio feature vector, M is a vector that gives the average of each audio feature, STD is a vector of standard deviation of each audio feature, and I_{norm} is the normalized feature vector. M and STD vectors are calculated for a given corpus.

In addition to the normalization, we also apply a weight vector W to balance the effect of each audio feature on the Euclidean distance calculation. The weights of MFCC dimensions are $1/13$ given that 13 MFCCs are calculated. The rest of the weight vector dimensions are set to 1. Hence, the total MFCC distances read as a timbre distance, and MFCC distances do not suppress the effect of remaining features on distance calculation. The SOM training input vector is,

$$I_{SOM} = I_{norm} * W \quad (3)$$

We have concluded on the equation 1, 2, and 3 after several trials with variety of corpora including electro-acoustic music, noise and glitch, IDM, and mainstream electronic music.

Musical Structure Generation with Statistical Sequence Modelling Algorithms

After the automatic creation of sonic latent space with SOMs, the question is, how do we model the organization of sound objects in time? ArchXXX learns the temporal musical structure by using a custom symbolic representation of compositions. This symbolic representation is a string of SOM nodes, and Figure 1 illustrates how the agent generates this string. First, ArchXXX labels each audio segment in a recording with its closest SOM node in the trained map. Second,

[2] http://forumnet.ircam.fr/product/mubu-en/

[3] http://ismm.ircam.fr/pipo/

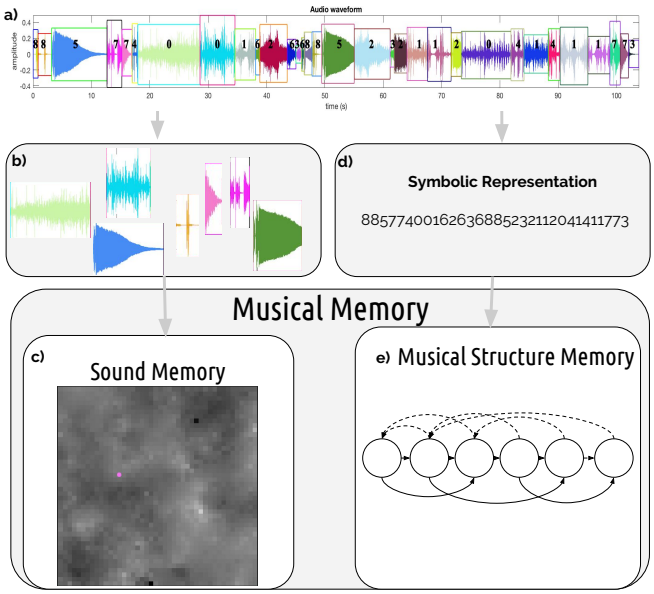

Figure 1: The training of ArchXXX: a) Segmentation b) Labelling the audio samples c) Sound memory where squares stand for SOM nodes that are a clusters of audio samples d) Creating a symbolic representation of the original song using the clusters indexes of audio samples e) statistical sequence model to learn temporal transitions.

ArchXXX uses the original order of the audio segments to create a string of SOM nodes. This procedure is repeated for each recording. Then, the agent learns the musical structure using these symbolic representations of recordings.

We investigated three algorithms for the musical structure modelling in ArchXXX: Variable-Markov Model Max-order variation (VMM-MO), Variable Markov Model Prediction by Partial Matching - C variation (VMM-PPM-C), and Factor Oracle. In the following, we explain how these algorithms work.

Markov Models are finite state machines that model patterns in discrete sequences using the Markov assumption. An Nth order Markov model assumes the following:

$$P(s_t|s_{t-1}, s_{t-2}, ..., s_1) = P(s_t|s_{t-1}, ..., s_{max(t-N,1)}) \quad (4)$$

Hence, Markov models are history dependent. The order of a Markov model indicates how many previous states to be considered to predict or generate the next state. Moreover, the frequency of transitions between Markov states determines the conditional probabilities of the transitions. Therefore, a Markov Model is a stochastic model represented as a directional graph.

Variable Markov models (VMMs) refer to a family of algorithms such as probabilistic finite automata, probabilistic suffix automata, prediction suffix trees, Lempel-Ziv 78, improved Lempel-Ziv (LZ-MS), Prediction by Partial Match (PPM), Factor Oracle, and the context tree weighting method [Ron *et al.*, 1996; Begleiter *et al.*, 2004]. VMM considers a varying number of previous states to predict the next state. VMMs are applied to a variety of musical tasks in the musical agents literature [Tatar and Pasquier, 2018, p. 25–30].

The log-loss indicates the performance of a compression algorithm where a lower log-loss implies better compression rates [Begleiter *et al.*, 2004]. However, a VMM model does not necessarily observe all possible sequences during the training. The probability of an unobserved sequence is zero. Hence, the log-loss of such sequence is infinite. This is known as the zero frequency problem in pattern matching and recognition. VMM variations handle zero frequency using an escape method.

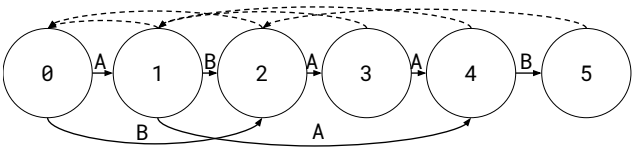

Figure 2: Factor Oracle generated using the sequence $ABAAB$.

The escape method of VMM-MO is as follows. When the model encounters a sequence that hasn't been observed during the training, the order decreases by one and the symbol at the most distant past is discarded. This processes is repeated until a known sequence is found. The model uses the found known sequence to predict the next symbol. In the most extreme case, the model decreases the order to zero, and predicts a symbol using the empty context. The max-order VMM has been previously shown to be successful for musical applications [Pachet, 2003; Pachet, 2004].

We implement a max-order VMM where the model is a list of transition matrices and count matrices stored as nested lists. Each nested list represents the transition matrix of a Markov Model of an order. The implementation is a Java code in MAX using the *max-mxj* framework.

The escape method of VMM-PPM introduces a probabilistic approach to the escape method of VMM-MO. Given a training sequence with length D, for each context s with length $k \leq D$, VMM-PPM partitions a total probability, $P_k(escape|s)$ between all symbols that does not appear after the context s. PPM allocates the remaining probability, $1 - P_k(escape|s)$ between the symbols that appear after the context s. The PPM variant is determined by how $P_k(escape|s)$ is calculated and how $1 - P_k(escape|s)$ is distributed amongst the symbols with non-zero counts [Begleiter *et al.*, 2004].

Several variants of VMM-PPM have been proposed in the literature. In particular, the escape mechanism of VMM-PPM-C is as follows. Given a maximum VMM order n, context s, and symbol σ,

$$P(\sigma|s) = \frac{f_\sigma}{L + M} \quad (5a)$$

$$P(escape|s) = \frac{M}{L + M} \quad (5b)$$

where M is the number of unique symbols in the alphabet, L is the sum of frequency counts of all symbols in s, and f_σ is the frequency count of a symbol σ. When the escape mechanism happens, PPM-C decreases the order by 1 and calculates the probabilities for the order $n - 1$.

Similar to the VMM learning, ArchXXX variations with Factor Oracle (FO) use the SOM string representation of a recording to train the FO. FO is a finite state automata that

represents substrings and patterns in a sequence, that is, at least all *factors* of a sequence (Figure 2). FO has three types of links: internal links, external links, and suffix links. Internal links are forward links between successive states. External links are forward links that jump longer than successive states. Suffix links are backward links that point the longest repeating factor in previous states. FO allows incremental learning, and learning is linear in time and space [Lefebvre and Lecroq, 2002]. [Assayag and Dubnov, 2004] compare IP, PSTs, and FOs for the symbolic sequences of music. [Assayag and Dubnov, 2004] conclude that FOs suit the best to satisfy incremental and fast online learning, time-bounded generation of musical sequences, and implementation of multi-attribute models to deal with the multi-dimensionality of music. Within the last two decades, many studies implemented FOs in musical agents [Assayag and Dubnov, 2004; Assayag *et al.*, 2006; Dubnov *et al.*, 2007; Lévy *et al.*, 2012; Nika and Chemillier, 2012; Surges and Dubnov, 2013; Nort *et al.*, 2013; Wang and Dubnov, 2014; Nika *et al.*, 2015; Nika *et al.*, 2017].

4 ArchXXX as a Partner, a Tool, and a Product

4.1 ArchXXX in Work X1 as Partner

The initial experiments with ArchXXX involved training the ML pipeline using the first author's musical improvisations recordings in noise music style. The author's familiarity with that content allowed a first person analysis on the ML pipeline, and the agent's musical capabilities and incapabilities. These first-person experiences with the musical agent evolved into design iterations of audio feature selections. The audio features included in the label vectors of audio samples are handpicked through a design process, and we first started with two affect features of eventfulness and pleasantness. We realized that using these two affect features, the system was unable to grasp the timbre information in the recordings although the eventfulness patterns in the training data was apparent. As the second step, we added a comprehensive timbre feature set, 13 MFCCs which is shown to outperform other audio features in the task of audio timbre matching [Yee-King, 2011]. This feature addition greatly improved the success of clustering and addressed the inconsistencies in the musical output. However, in rare cases, we observed that some clusters included samples that we perceived to fit in two different, but closely related clusters. Hence, in the third step, we included perceptual spectral decrease feature to further improve the SOM clustering. In the three initial iterations of audio feature selections, we mainly focused on IDM and noise music where the loudness variations and the dynamic range was low. Our experiments with electroacoustic corpus indicated that the agent's musical output was jumping between various loudness levels in a random manner. Hence, in the fourth step, we added the loudness audio feature to the set. In the last step, we added duration of samples to the feature set to decrease duration variations within SOM clusters.

The Work X1 emerged within these design iterations on audio feature selections that mattered in musical qualities of the agent's output. In this first artwork output of ArchXXX, the statistical sequence generation algorithm was VMM-MO, which is the same algorithm used in a well-known musical AI work titled Continuator [Pachet, 2003]. All variants of VMM allow realtime interactive applications, since the predictions are carried out for any given input sequence. The agent was trained on a noise free improvisation album where the first author performed alone. The audio recordings approximately sums to one hour in duration. In this first performance, the audio input of the agent was both the human performer and the agents audio output (see Figure 3). Hence, the agent was listening in realtime to the to musical performance as a whole. The performance setup included two speakers, where the machine's and human performer's audio outputs were on different speakers. Both the machine and the human performer was working with similar sonic material and close in timbre to each other, since the agent was trained on the audio recordings where human performer improvised alone using the same digital musical instruments. The two-speaker performance setup allowed a presentation to differentiate machine sonic actions than the actions of the human performer.

Consequently, the presentation of this first artwork manifested the ArchXXX based musical agent as a performer that exchanged sonic actions with a human performer. This type of presentation positions the AI system as "the other", and in AI-ethics, the issues related to such contextualization of AI systems has been discussed previously (see Section 5). From the spectators' perspective, the connection between the human and the machine in this artwork is only apparent in the similarity of musical timbre in their sonic actions. We argue that such similarity in sonic domain created an interaction space that encompassed the sonic world of both performers. That interaction space enables the spectators to recognize realtime interactions between the machine and the human through the similarities in timbre and sonic actions.

4.2 ArchXXX in Work X2 No.1 as a Product in an Interactive Installation

Work X2 is a series of artworks that explores the theme of plastic pollution while emphasizing how plastic material appear ubiquitously in almost all aspects of urban daily life. The first edition of this work was an interactive artwork in audio-visual domain. The interaction between the audience and the artwork utilized a well-known approach in interactive arts, called "mirroring", where the interaction emerges from a process in which the audience body is captured and processed as a material in the artwork. In Work X2 No.1, the audience body is captured using a kinect camera. The silhouette of the body is passes to a pixel sorting algorithm that manipulates a realtime video stream. The pixelation of the video initiate a playful interaction with the audience and the artwork, while touching on the destructive affect of plastic material on daily life. The video stream is urban scenery that is stylized using a Deep Learning algorithm [Johnson *et al.*, 2016; Wang *et al.*, 2020; Zou *et al.*, 2021].

Preliminary to the production of Work X2, we updated the ArchXXX by introducing a sample dataset to the architecture. ArchXXX has a unique ability to create a symbolic representation of any recording. The symbolic representation is created through first creating a latent sound sample space using self-organizing maps, then using that latent space to create a symbolic representation where the symbols represent cluster numbers (see Figure 1). In this update of ArchXXX, the ar-

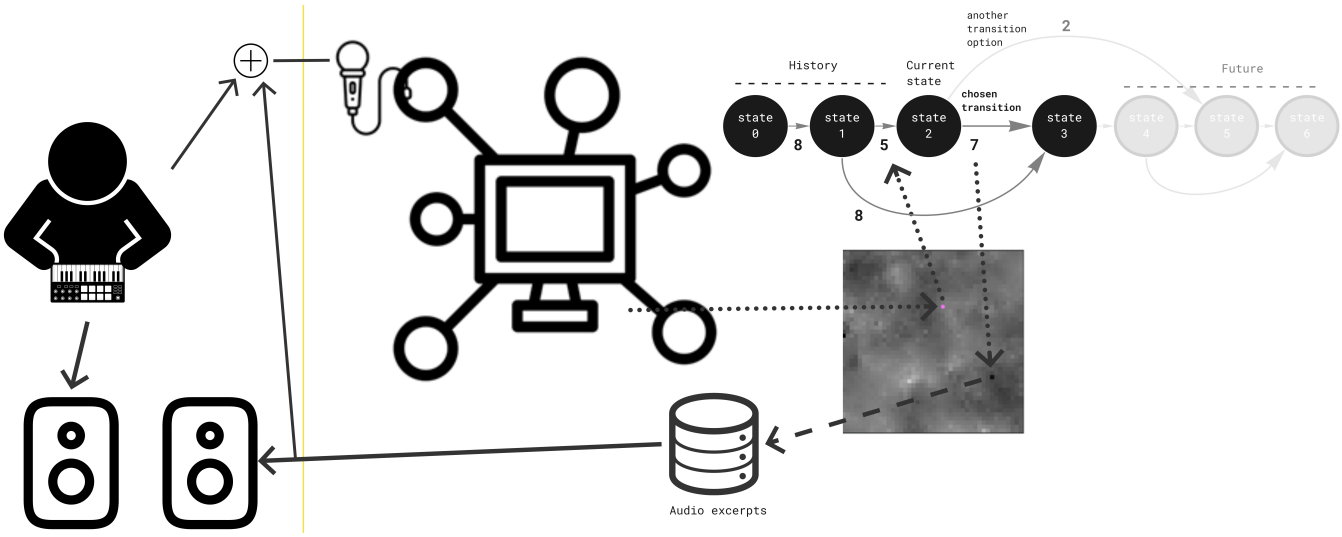

Figure 3: System overview of ArchXXX utilized as a partner in musical performance

Figure 4: The artwork X2 No.1 at the exhibition EX1.

chitecture includes two types of dataset: a recording dataset, composition dataset, and a sample dataset. The recording dataset is processed through the full pipeline where the audio files are segmented and thumbnailed using audio feature extraction. The sample dataset bypasses the segmentation since the files are ideally single sonic gestures. The segments in the recording dataset and the sound samples in the sample dataset are gathered together to train the self-organizing map. The sounds in those two libraries are also added to the playback library that the agent utilizes in the inference. The trained latent sound space model is used to create a symbolic representation of audio recordings in the recording dataset and the composition dataset. Notice that the sound segments in the composition dataset recordings are not added to the playback library and the recordings in the composition dataset is utilized only for extracting the temporal musical form in those recordings. We further discuss the future copyright challenges regarding this in Section 5.

The musical agent in Work X2 utilizes ArchXXX in self-listening mode. In this mode, the pipeline acts as a single performer which transforms the interactive agent into a generative music system. In the self-listening mode additional statistical sequence models that works as state machines, such as Factor Oracle, are possible to be exchanged with the VMM which affords an exploration of different statistical sequence generation algorithms within a generative music system. When an audience steps towards the camera in Work X2, the sound output increased in volume which invites the audience into the artwork by surrounding the audience with a sonic environment. Although the musical agent's direct involvement in the audience interaction is minimal, the sonic possibilities of the agent was the initiation point for the aesthetics explorations in the bringing out of this work (CITEX).

The sequence models in this iteration of ArchXXX are trained on the symbolic representation of a single recording in the dataset, and the training time is less than half a second. Using the same latent audio space, it is possible to exchange the sequence models during the presentation of the generative music system. This gives a flexibility in presenting the trained ArchXXX models as a generative music system with different scenes covering a variety of aesthetics. Different than sonic interactions in Work X1 where ArchXXX was positioned as a partner in musical performance, the scene changes in Work X2 comes out as separate audiovisual environments with their own sonic space, which manifests the generative music system based on ArchXXX as a *product*, a gestalt that is finalized in its aesthetics.

4.3 ArchXXX in Work X2 No.2 as a Tool in Musical Production

The second iteration of Work X2 is a live audiovisual artwork where the audience body is replaced with a contemporary dancer. The interaction between the dancer and generative visuals are more explicit in the visual pixelation effect, whereas the interaction between the dance and the music is navigated within the multimodal performance. On the musical side, ArchXXX is deployed as a generative music system which is accompanied by a human musician. In this performance setup, the ArchXXX does not listen to the human mu-

sician as in the case of Work X1, and this decision positions the musical XYZ of ArchXXX model as a *product*, rather than a *partner*. The musical output of the generative system is positioned in the role of a sonic background, which depicts a sonic space for the human performer build on. The gestalt, which is the multimodal performance in this case, inherits the generative product, which is a trained model of ArchXXX.

In comparison to the gestalt, the processes in the production of Work X2 No.2 utilizes the ArchXXX as tool to navigate an exploration of aesthetics. After being trained on a dataset, ArchXXX includes a single latent sound space of audio segments, and a number statistical sequence models where each sequence model is trained on the custom symbolic representation of single recording. Each sequence model uses a subset of audio segment clusters in the latent sound space, which constraints the musical output to a specific aesthetic. By changing the statistical sequence model, the artist can navigate various sonic aesthetics with ease, where each sequence model is a generative music system of its own accord. In that artistic production process, the functionality of ArchXXX is a *tool* that helps the artist to explore different aesthetics.

5 Discussions

AI ethics researcher, Bryson [Ars Electronica, 2017] has raised a precaution on how current contextualization of AI present these systems as independent actors of their own. Bryson lays down moral hazards, distractions from real threats in public discussions, ethical coherence, legal lacuna that can emerge from contextualizing AI systems as the *other*. In the presentations of AI as the *other*, the power relationships between the machine and its makers are often clouded. In the current discussions of Artificial Intelligence, involved actors in AI system creation such as system designers, dataset curators, and hardware makers are not necessarily concealed. In the Work X1, the audience was informed on the mechanisms of musical agent and its training processes; however, the actors of ArchXXX is not revealed to the spectators during the musical performance in its depiction since the interaction scenario manifested the machine as a performer of its own. More research into the power mechanisms in musical AI systems can reveal the hierarchies and structures in designing and processes of those systems, which may not be always clear to the spectator point of view.

In AI and ML research, the emphasis is often put on the *gestalt* instead of the *process*. It is common the the final product that is either an architecture or a trained model, is validated or evaluated while the processes are often overlooked. In artistic research, we argue that the processes play a important role in how the aesthetics evolve into a final artwork. Those processes in aesthetics goes beyond a single artwork and often encompasses the full career of an artist. An investigation of long-term processes in artistic practices using advance technologies can be beneficial to both artistic communities in Art and Technology, and academic communities researching interaction with the technology.

In Section 4.2, we mentioned that the current version of ArchXXX which has a capability to extract and model temporal form in any audio recording, which allows user to model a musical composition using a sound sample library and produce a generative music system based on the musical form of a recording. The ability to model musical form and generate audio recordings using different sound samples challenges the current ethics and regulations on copyright in music and sound practices. Currently, the copyright of such material is built on the audio similarity or similarities in conventional musical notation. While ArchXXX was modelling the musical form, one can question if the trained sequence model is a representation of the musical style, or the musical form in the recording. That is, how do we differentiate between the musical form of a single composition and a musical style? Where does the copyright of a composition and a musical style begin? For example, many electronic dance music genres involves drum sounds that are specific to a subgenre, such as TR 808 sounds in house music and TR 909 sounds in techno. In that example, the close similarities between drum sounds are one of the main factors that indicate the musical subgenre. These examples amongst others have raised the question on what a musical style is [Dannenberg, 2010; Collins *et al.*, 2016], and such definition may be specific to culture that the musical style situates in.

Many actors are involved in musical AI systems in addition to the artists, musicians, and creative practitioners; actors such as AI system architecture designers and developers, owners and creators of data, curators who collects data to create datasets, organizations that bring forward frameworks for AI algorithms such as (PyTorch and TensorFlow), actors involved in hardware systems and so on. A stakeholder analysis is a known method to discover involved participants of a phenomena. A discovery of all actors of musical AI systems can create the foundations for revealing all types of interactivity with musical AI systems and their real-world applications.

6 Conclusions and Future Work

By investigating the interactivity in the artistic applications of ArchXXX, we took a first step into revealing different roles of musical AI in artworks and artistic practices. This initial thematic analysis revealed three distinct categories of musical AI taking the role of a partner, a product, and a tool in their integration into artworks. The thematic analysis was through first-person experiences in a long-term artistic processes and the gestalt presented as a result of those processes. The design processes in the technology development, the aesthetics explorations, and artwork production span multiple artworks, which enabled us the highlight the different roles that the same musical AI system can take depending on how its is integrated within an artwork. This initial step highlights the necessity in further research to reveal the various dimensions of interactivity with musical AI architectures.

Ethical Statement

The system design in this work built on the musical theories in electroacoustic music, such as the notion of "organized sound". We acknowledge that the electroacoustic music theories that the system design is based on, are mainly founded on the track of the Western music theories, and the theories in these disciplines have yet to reach a diversity that encompasses a variety of musical cultures and their theories across the globe.

Notes

ArchXXX is not the system's name, rather a spaceholder to ensure double-blind review. Similarly, Work X1 and X2 are placeholders for the names of two artworks, which are reducted for the reviewing. CITEY is a paper by the first author published in a conference. CITEX refers to a conference paper that is currently in review.

The supplementary material is available at https://drive.google.com/drive/folders/1_bLodG9wCiHMZ6Gl226g7CNdSe9iC_c7?usp=share_link

References

[Ars Electronica, 2017] Ars Electronica. Ars Electronica 2017 - Symposium I - Ethics, Philosophy and Spirituality (EN), September 2017.

[Assayag and Dubnov, 2004] G. Assayag and S. Dubnov. Using Factor Oracles for Machine Improvisation. *Soft Computing*, 8(9):604–610, August 2004.

[Assayag *et al.*, 2006] Gérard Assayag, Georges Bloch, Marc Chemillier, Arshia Cont, and Shlomo Dubnov. Omax brothers: a dynamic topology of agents for improvization learning. In *Proceedings of the 1st ACM Workshop on Audio and Music Computing Multimedia*, pages 125–132. ACM Press, 2006.

[Begleiter *et al.*, 2004] Ron Begleiter, Ran El-Yaniv, and Golan Yona. On prediction using variable order Markov models. *Journal of Artificial Intelligence Research*, 22:385–421, 2004.

[Briot *et al.*, 2017] Jean-Pierre Briot, Gaëtan Hadjeres, and François Pachet. Deep Learning Techniques for Music Generation-A Survey. *arXiv preprint arXiv:1709.01620*, 2017.

[Bødker, 2006] Susanne Bødker. When second wave HCI meets third wave challenges. In *Proceedings of the 4th Nordic conference on Human-computer interaction: changing roles*, NordiCHI '06, pages 1–8, New York, NY, USA, October 2006. Association for Computing Machinery.

[Bødker, 2015] Susanne Bødker. Third-wave HCI, 10 years later—participation and sharing. *Interactions*, 22(5):24–31, August 2015.

[Carnovalini and Rodà, 2020] Filippo Carnovalini and Antonio Rodà. Computational Creativity and Music Generation Systems: An Introduction to the State of the Art. *Frontiers in Artificial Intelligence*, 3:14, April 2020.

[Clarke and Braun, 2017] Victoria Clarke and Virginia Braun. Thematic analysis. *The Journal of Positive Psychology*, 12(3):297–298, May 2017. Publisher: Routledge _eprint: https://doi.org/10.1080/17439760.2016.1262613.

[Collins *et al.*, 2016] Tom Collins, Robin Laney, Alistair Willis, and Paul H. Garthwaite. Developing and evaluating computational models of musical style. *Artificial Intelligence for Engineering Design, Analysis and Manufacturing*, 30(01):16–43, February 2016.

[Cross, 2006] Nigel Cross, editor. *Designerly Ways of Knowing*. Springer, London, 2006.

[Dannenberg, 2010] Roger B. Dannenberg. Style in Music. In Shlomo Argamon, Kevin Burns, and Shlomo Dubnov, editors, *The Structure of Style*, pages 45–57. Springer Berlin Heidelberg, 2010.

[Desjardins *et al.*, 2021] Audrey Desjardins, Oscar Tomico, Andrés Lucero, Marta E. Cecchinato, and Carman Neustaedter. Introduction to the special issue on first-person methods in hci. *ACM Trans. Comput.-Hum. Interact.*, 28(6), dec 2021.

[Dubnov *et al.*, 2007] Shlomo Dubnov, Gerard Assayag, and Arshia Cont. Audio Oracle: A New Algorithm for Fast Learning of Audio Structures. In *Proceedings of International Computer Music Conference*, 2007.

[Ellis and Bochner, 2000] Carolyn S. Ellis and Arthur Bochner. Autoethnography, Personal Narrative, Reflexivity: Researcher as Subject. In *The Handbook of Qualitative Research*. Sage, 2000.

[Fernández Rodríguez and Vico Vela, 2013] Jose David Fernández Rodríguez and Francisco José Vico Vela. AI Methods in Algorithmic Composition: A Comprehensive Survey. *Journal of Artificial Intelligene Research*, 48:513–582, 2013.

[Filimowicz and Tzankova, 2018] Michael Filimowicz and Veronika Tzankova. Introduction | New Directions in Third Wave HCI. In Michael Filimowicz and Veronika Tzankova, editors, *New Directions in Third Wave Human-Computer Interaction: Volume 1 - Technologies*, Human–Computer Interaction Series, pages 1–10. Springer International Publishing, Cham, 2018.

[Frayling, 1994] Christopher Frayling. Research in Art and Design (Royal College of Art Research Papers, Vol 1, No 1, 1993/4). *Royal College of Art Research Papers*, 1(1), 1994.

[Herremans *et al.*, 2017] Dorien Herremans, Ching-Hua Chuan, and Elaine Chew. A Functional Taxonomy of Music Generation Systems. *ACM Computing Surveys*, 50(5):1–30, September 2017.

[Höök *et al.*, 2021] Kristina Höök, Steve Benford, Paul Tennent, Vasiliki Tsaknaki, Miquel Alfaras, Juan Martinez Avila, Christine Li, Joseph Marshall, Claudia Daudén Roquet, Pedro Sanches, Anna Ståhl, Muhammad Umair, Charles Windlin, and Feng Zhou. Unpacking non-dualistic design: The soma design case. *ACM Trans. Comput.-Hum. Interact.*, 28(6), nov 2021.

[Ihde, 1990] Don Ihde. *Technology and the Lifeworld: From Garden to Earth*. Indiana University Press, 1990.

[Johnson *et al.*, 2016] Justin Johnson, Alexandre Alahi, and Li Fei-Fei. Perceptual Losses for Real-Time Style Transfer and Super-Resolution, March 2016. arXiv:1603.08155 [cs].

[Kiger and Varpio, 2020] Michelle E. Kiger and Lara Varpio. Thematic analysis of qualitative data: AMEE Guide No. 131. *Medical Teacher*, 42(8):846–854, August 2020.

[Kohonen, 1982] Teuvo Kohonen. Self-organized formation of topologically correct feature maps. *Biological cybernetics*, 43(1):59–69, 1982.

[Kohonen, 1998] Teuvo Kohonen. The self-organizing map. *Neurocomputing*, 21(1–3):1–6, November 1998.

[Krogh and Koskinen, 2015] Peter Krogh and Ilpo Koskinen. Design accountability: When design research entangles theory and practice. *International Journal of Design*, 9:121–127, 04 2015.

[Lefebvre and Lecroq, 2002] Arnaud Lefebvre and Thierry Lecroq. A Heuristic For Computing Repeats With A Factor Oracle: Application To Biological Sequences. *International Journal of Computer Mathematics*, 79(12):1303–1315, January 2002.

[Lévy *et al.*, 2012] Benjamin Lévy, Georges Bloch, and Gérard Assayag. OMaxist dialectics. In *Proceedings of the International Conference on New Interfaces for Musical Expression (NIME)*, pages 137–140, 2012.

[Lucero *et al.*, 2021] Andrés Lucero, Audrey Desjardins, and Carman Neustaedter. Longitudinal First-Person HCI Research Methods. In Evangelos Karapanos, Jens Gerken, Jesper Kjeldskov, and Mikael B. Skov, editors, *Advances in Longitudinal HCI Research*, Human–Computer Interaction Series, pages 79–99. Springer International Publishing, Cham, 2021.

[McCarthy and Wright, 2007] John McCarthy and Peter Wright. *Technology as Experience*. 2007.

[Neustaedter and Sengers, 2012] Carman Neustaedter and Phoebe Sengers. Autobiographical design in hci research: Designing and learning through use-it-yourself. In *Proceedings of the Designing Interactive Systems Conference*, DIS '12, page 514–523, New York, NY, USA, 2012. Association for Computing Machinery.

[Nierhaus, 2009] Gerhard Nierhaus. *Algorithmic Composition: Paradigms of Automated Music Generation*. Springer Science & Business Media, August 2009.

[Nika and Chemillier, 2012] Jérôme Nika and Marc Chemillier. Improtek: integrating harmonic controls into improvisation in the filiation of OMax. In *International Computer Music Conference (ICMC)*, pages 180–187, 2012.

[Nika *et al.*, 2015] Jérôme Nika, Dimitri Bouche, Jean Bresson, Marc Chemillier, and Gérard Assayag. Guided improvisation as dynamic calls to an offline model. In *Sound and Music Computing (SMC)*, Maynooth, Ireland, July 2015.

[Nika *et al.*, 2017] Jérôme Nika, Marc Chemillier, and Gérard Assayag. ImproteK: Introducing Scenarios into Human-Computer Music Improvisation. *Computers in Entertainment*, 14(2):1–27, January 2017.

[Nort *et al.*, 2013] Doug Van Nort, Pauline Oliveros, and Jonas Braasch. Electro/Acoustic Improvisation and Deeply Listening Machines. *Journal of New Music Research*, 42(4):303–324, December 2013.

[Pachet, 2003] Francois Pachet. The continuator: Musical interaction with style. *Journal of New Music Research*, 32(3):333–341, 2003.

[Pachet, 2004] François Pachet. Beyond the cybernetic jam fantasy: The continuator. *Computer Graphics and Applications, IEEE*, 24(1):31–35, 2004.

[Redström, 2017] Johan Redström. *Making Design Theory*. The MIT Press, 2017.

[Ron *et al.*, 1996] Dana Ron, Yoram Singer, and Naftali Tishby. The power of amnesia: Learning probabilistic automata with variable memory length. *Machine learning*, 25(2-3):117–149, 1996.

[Rosenberger and Verbeek, 2015] Rosenberger and Peter-Paul Verbeek. *Postphenomenological Investigations: Essays on Human–Technology Relations*. Lexington Books, May 2015. Google-Books-ID: vpqcCQAAQBAJ.

[Schnell *et al.*, 2009] Norbert Schnell, Axel Röbel, Diemo Schwarz, Geoffroy Peeters, and Riccardo Borghesi. MuBu and friends–assembling tools for content based real-time interactive audio processing in Max/MSP. In *Proceedings of International Computer Music Conference (ICMC)*, 2009.

[Surges and Dubnov, 2013] Greg Surges and Shlomo Dubnov. Feature selection and composition using PyOracle. In *Ninth Artificial Intelligence and Interactive Digital Entertainment Conference*, 2013.

[Svanæs and Barkhuus, 2020] Dag Svanæs and Louise Barkhuus. The designer's body as resource in design: Exploring combinations of point-of-view and tense. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, CHI '20, page 1–13, New York, NY, USA, 2020. Association for Computing Machinery.

[Tatar and Pasquier, 2018] Kıvanç Tatar and Philippe Pasquier. Musical agents: A typology and state of the art towards Musical Metacreation. *Journal of New Music Research*, 48(1):1–50, September 2018.

[Tatar and Pasquier, 2019] Kıvanç Tatar and Philippe Pasquier. Musical agents: A typology and state of the art towards Musical Metacreation. *Journal of New Music Research*, 48(1):56–105, 2019.

[Varela and Shear, 1999] Francisco Varela and Jonathan Shear. First-person methodologies: What, why, how? *Journal of Consciousness Studies*, 6(2-3):1–14, 1999.

[Verbeek, 2008] Peter-Paul Verbeek. Cyborg intentionality: Rethinking the phenomenology of human–technology relations. *Phenomenology and the Cognitive Sciences*, 7(3):387–395, September 2008.

[Wang and Dubnov, 2014] Cheng-i Wang and Shlomo Dubnov. Guided music synthesis with variable markov oracle. In *The 3rd International Workshop on Musical Metacreation, 10th Artificial Intelligence and Interactive Digital Entertainment Conference*, 2014.

[Wang *et al.*, 2020] Wenjing Wang, Shuai Yang, Jizheng Xu, and Jiaying Liu. Consistent Video Style Transfer via Relaxation and Regularization. *IEEE Transactions on Image Processing*, 29:9125–9139, 2020. Conference Name: IEEE Transactions on Image Processing.

[Williams *et al.*, 2015] Duncan Williams, Alexis Kirke, Eduardo R Miranda, Etienne Roesch, Ian Daly, and Slawomir Nasuto. Investigating affect in algorithmic composition systems. *Psychology of Music*, 43(6):831–854, November 2015.

[Yee-King, 2011] Matthew John Yee-King. *Automatic sound synthesizer programming: techniques and applications*. PhD thesis, University of Sussex, 2011.

[Zimmerman *et al.*, 2007] John Zimmerman, Jodi Forlizzi, and Shelley Evenson. Research through design as a method for interaction design research in hci. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, CHI '07, page 493–502, New York, NY, USA, 2007. Association for Computing Machinery.

[Zou *et al.*, 2021] Zhengxia Zou, Tianyang Shi, Shuang Qiu, Yi Yuan, and Zhenwei Shi. Stylized Neural Painting. In *Proceedings of Conference on Computer Vision and Pattern Recognition (CVPR) 2021*, 2021.